# Refund processing

Architecture overview. Readable node and connection sheets follow.



# Refund processing

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

## Owner requests refund

Payments UI verifies provider payment before refunding.

src/app/admin/(control-center)/payments/page.tsx:1

Next -> #2 (p2)

#02 / HTTP Request / Implemented

## POST admin payments

Owner authorization, origin, rate limit and refund input checks.

src/app/api/admin/payments/route.ts:50

Next -> #3 (p2)

#03 / IF / Switch / Implemented

## Refundable captured payment?

Verify provider capture, currency, amount, remaining refundable balance and idempotent refund identity.

src/server/razorpay-refunds.ts:100

Yes / accepted -> #4 (p2); No / rejected -> #10 (p4)

#04 / Database Query / Implemented

## Record Processing refund

Embedded orders.refunds[] and audit entry record the intent before provider execution.

src/server/razorpay-refunds.ts:408

Next -> #5 (p3)

# Refund processing

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / External API / Implemented

## Razorpay Refunds API

Gateway may report processed, failed or still processing; ambiguous responses remain unresolved. Runtime configuration and connectivity are Unverified.

src/server/razorpay-refunds.ts:430

Next -> #6 (p3)

#06 / Webhook / Implemented

## Signed refund webhook

Verified Razorpay refund event reaches processRefundWebhook.

src/app/api/webhooks/[provider]/route.ts:13

Next -> #7 (p3)

#07 / Backend Service / Implemented

## Reconcile remote refund

Refund identity, order association, amount and event status are validated.

src/server/razorpay-refunds.ts:561

Next -> #8 (p3)

#08 / Database Query / Implemented

## Refunded / Failed / Processing

Persist only provider-confirmed status; audit and cancellation reconciliation follow.

src/server/razorpay-refunds.ts:284

Next -> #9 (p4)

# Refund processing

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Response / Implemented

## Provider refund status

Owner and account views use recorded provider refund status.

src/app/admin/(control-center)/payments/page.tsx:1

END: no outgoing connection

#10 / Error Handler / Implemented

## Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #11 (p4)

#11 / Response / Implemented

## Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

# Refund processing

Complete directed connection register / no omitted success, failure or proposed edges

#01 Owner requests refund -> #02 POST admin payments | Next

#02 POST admin payments -> #03 Refundable captured payment? | Next

#03 Refundable captured payment? -> #04 Record Processing refund | Yes / accepted

#04 Record Processing refund -> #05 Razorpay Refunds API | Next

#05 Razorpay Refunds API -> #06 Signed refund webhook | Next

#06 Signed refund webhook -> #07 Reconcile remote refund | Next

#07 Reconcile remote refund -> #08 Refunded / Failed / Processing | Next

#08 Refunded / Failed / Processing -> #09 Provider refund status | Next

#03 Refundable captured payment? -> #10 Request failed | No / rejected | FAILURE

#10 Request failed -> #11 Show error / retry | Next | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: Manual returns approval is separate from gateway refund execution. Automated PayPal and Stripe refund execution was not found; a future implementation is Planned.